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Chapter 21

Routing with EIGRP

The ICND2 exam tests your knowledge of different routing protocols, and one of the protocols you will receive questions on is the EIGRP routing protocol. Study the details of EIGRP, and then try the questions in this chapter to test your knowledge level of Cisco's hybrid routing protocol!

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Understanding EIGRP
- Configuring EIGRP
- Troubleshooting EIGRP

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- Know the details of EIGRP, such as the fact that it is a hybrid routing protocol that uses some distance vector features and link state routing protocol features. EIGRP has an administrative distance and is a Cisco proprietary protocol.
- Know the difference between a successor route and a feasible successor route. EIGRP stores the best route to a destination, known as a successor route, in the routing table. All routes are stored in the topology table as well, including backup routes to the successor route. These backup routes are known as feasible successor routes.
- When the successor route fails, the feasible successor route immediately becomes the successor and a new feasible successor route is elected. Having the backup (feasible successor) stored in the topology table reduces convergence time because it is available immediately.
- Know the commands to configure EIGRP. You use the router eigrp <AS #> command to enable EIGRP. You then use the network command with the network ID and the wildcard mask to enable EIGRP routing on that network. Know how to calculate the wildcard mask. The quickest method is to take the subnet mask away from 255.255.255.255. For example, if a subnet mask is 255.255.255.192, then the wildcard mask for that network is 0.0.0.63.
- Know your commands to troubleshoot EIGRP. Some commands you should be familiar with are `show ip route`, `show ip eigrp neighbors`, `show ip eigrptopology`, and `show ip protocols`. Know which commands display what information: That can help you troubleshoot in scenario-based questions.

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Understanding EIGRP

852–872 Choose the best answer(s).

852. Which of the following is used when the successor route fails?

(A) Alternate route

(B) Feasible successor route

(C) BDR

(D) DR

853. What is the purpose of the feasible successor route?

(A) It is a backup route located in the routing table.

(B) It is the primary route located in the routing table.

(C) It is the primary route located in the topology table.

(D) It is a backup route located in the topology table.

854. Which of the following is true in regards to the autonomous system (AS) number with EIGRP?

(A) It should be unique on each router.

(B) It must match on all routers that you want to share routing information.

(C) It must match the value of the last octet in the IP address of the WAN interface.

(D) It represents the hop count to the next network.

855. What happens when EIGRP has two paths with equal metrics to the same destination?

(A) It uses the first path.

(B) It uses the shortest path.

(C) It load balances between the two paths.

(D) It uses the last path.

856. In regards to the commands shown below, what does the 10 represent?

```
NY-R1(config)#router eigrp 10
NY-R1(config-router)#network 11.0.0.0
```

(A) The maximum hop count

(B) The maximum equal paths

(C) The autonomous system

(D) The administrative distance

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857. The EIGRP algorithm looks at all the routes in the topology table and chooses the best route to a destination. The best route is known as the _____.

(A) Feasible successor route

(B) Successor route

(C) Designated route

(D) Backup designated route

858. Which table does the successor route exist in? (select two)

(A) Topology table

(B) Neighbors table

(C) Routing table

(D) MAC table

859. EIGRP is considered which of the following?

(A) A hybrid protocol

(B) A link state protocol

(C) A distance vector protocol

(D) A layer-2 protocol

860. What is the administrative distance of EIGRP?

(A) 150

(B) 120

(C) 110

(D) 90

861. Which table does the feasible successor route exist in?

(A) Neighbors table

(B) Routing table

(C) Topology table

(D) MAC table

862. A router has learned of three routes to the same destination. The first is from RIPv2 with a metric of 6, the second route is from OSPF with a metric of 10, and the third route is from EIGRP with a metric of 2172416. Which route will be used?

(A) The RIPv2 route

(B) The OSPF route

(C) All of them

(D) The EIGRP route

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863. What is the benefit of having a feasible successor route with EIGRP?

(A) Load balancing

(B) Quicker convergence

(C) Less hops

(D) More hops

864. With EIGRP, which of the following is also known as the routing domain ID?

(A) Router-ID

(B) BDR ID

(C) Metric value

(D) Autonomous system number

865. Which of the following describes a feasible successor?

(A) Backup route stored in topology table

(B) Primary route stored in topology table

(C) Backup route stored in routing table

(D) Primary route stored in routing table

866. Which of the following are true in regards to EIGRP? (Select two.)

(A) Is vendor specific

(B) Uses cost as the metric

(C) Has a default administrative distance of 90

(D) Elects a DR and BDR

867. Which of the following is true of the EIGRP successor route? (Select two.)

(A) Successor routes are backup routes to feasible successor routes.

(B) Successor routes forward traffic.

(C) May have a backup in a feasible successor route.

(D) Successor routes are used when a feasible successor route fails.

868. Which of the following describes a successor route?

(A) Backup route stored in topology table

(B) Primary route stored in neighbors table

(C) Backup route stored in routing table

(D) Primary route stored in routing table

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869. Which of the following does EIGRP use as its metrics by default? (Select two.)

(A) Hop count

(B) Cost

(C) Load

(D) Bandwidth

(E) Delay

870. Which of the following are true regarding successor routes? (Select three.)

(A) They are the backup to a feasible successor route.

(B) They are backed by feasible successor routes.

(C) They are stored in the neighbors table.

(D) They are used to forward traffic to the destination.

(E) They are stored in the routing table.

871. What does an EIGRP router do if its successor route is active but there is no feasible successor route?

(A) Elects a DR

(B) The router sends a multicast message to all adjacent neighbors.

(C) Elects a BDR

(D) Broadcast a message looking for a feasible successor

872. When looking at the EIGRP topology, what code is used to indicate that there are no changes in the EIGRP topology?

(A) Active

(B) Passive

(C) Reply status

(D) Query

Configuring EIGRP

873–879 Choose the best answer(s).

873. What command is used to start EIGRP on your router?

(A) R1(config)# `start eigrp 5`

(B) R1(config)# `start router eigrp`

(C) R1(config)# `router eigrp 5`

(D) R1(config)# `router eigrp`

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874. You wish to disable auto summarization with EIGRP. What command would you use?

(A) R1(config)# `no auto-summary`

(B) R1# `no auto-summary`

(C) R1(config-router)# `no auto-summary`

(D) R1(config-router)# `disable auto-summary`

875. You are configuring router R1 and wish to enable EIGRP on all interfaces for the 11.0.0.0 network and 12.0.0.0 network. What command would you use?

(A) Use the following commands:

```
R1(config)#router eigrp
R1(config-router)#network 11.0.0.0
R1(config-router)#network 12.0.0.0
```

(B) Use the following commands:

```
R1(config)#router eigrp 10
R1(config-router)#network 11.0.0.0 255.0.0.0
R1(config-router)#network 12.0.0.0 255.0.0.0
```

(C) Use the following commands:

```
R1(config)#router eigrp 10
R1(config-router)#network 11.0.0.0 0.255.255.255
R1(config-router)#network 12.0.0.0 0.255.255.255
```

(D) Use the following command:

```
R1(config)#router eigrp 11.0.0.0,12.0.0.0
```

876. You wish to change the administrative distance of EIGRP. What command would you use?

(A) R1(config)#`distance 35 35`

(B) R1(config-router)#`admin-distance 35 35`

(C) R1(config-router)#`distance 35 35`

(D) R1(config-router)#`distance eigrp 35 35`

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877. You have configured three routers with the configuration shown below. The routers are not exchanging routing updates. What should you do?

```
NY-R1(config)#router eigrp 10
NY-R1(config-router)#network 11.0.0.0
BOS-R1(config)#router eigrp 20
BOS-R1(config-router)#network 12.0.0.0
TOR-R1(config)#router eigrp 30
TOR-R1(config-router)#network 13.0.0.0
```

(A) Configure different AS on all routers.

(B) Configure the same AS on all routers.

(C) Promote the feasible successor.

(D) Force election of a new DR.

878. You have started the EIGRP process. What command would you use to enable EIGRP for network 192.168.3.64/27?

(A) network 192.168.3.64 255.255.255.0

(B) network 192.168.3.64 0.0.0.255

(C) network 192.168.3.64 0.0.0.31

(D) network 192.168.3.64 0.0.0.0

879. You have started the EIGRP process. What command would you use to enable EIGRP for network 172.16.16.0/20?

(A) `network 172.16.0.0 0.0.255.255`

(B) `network 172.16.16.0 0.0.255.255`

(C) `network 172.16.16.0 255.255.240.0`

(D) `network 172.16.16.0 0.0.15.255`

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Troubleshooting EIGRP

880–889 Choose the best answer(s).

880. Looking at the figure below, how many packets are waiting to be delivered to the EIGRP neighbor?

```
R1#show ip eigrp neighbors
IP-EIGRP neighbors for process 10
H   Address         Interface       Hold Uptime    SRTT   RTO   Q   Seq
                                    (sec)          (ms)        Cnt Num
0   12.0.0.2        Se0/3/0         10   00:03:11  40     1000  0   4
```

(A) 40

(B) 0

(C) 10

(D) 4

881. You have configured EIGRP on router R1 and would like to configure router R2 for the same AS. Looking at the figure below, what AS is R1 using?

```
R1#show ip protocols

Routing Protocol is "eigrp  10 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
  EIGRP maximum hopcount 100
  EIGRP maximum metric variance 1
Redistributing: eigrp 10
  Automatic network summarization is in effect
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
     11.0.0.0
     12.0.0.0
  Routing Information Sources:
    Gateway         Distance      Last Update
    12.0.0.2        90            99622
  Distance: internal 90 external 170
```

(A) 10

(B) 4

(C) 90

(D) 100

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882. What command was used in the figure below to display the output?

```
IP-EIGRP neighbors for process 10
H   Address         Interface       Hold Uptime    SRTT   RTO   Q   Seq
                                    (sec)          (ms)        Cnt Num
0   12.0.0.2        Se0/3/0         10   00:03:11  40     1000  0   4
```

(A) `show ip eigrp neighbors`

(B) `show ip eigrp topology`

(C) `show ip route`

(D) `show ip eigrp route`

883. You would like to view the IP addresses of neighboring devices with EIGRP. What command should you use?

(A) `show ip eigrp interfaces`

(B) `show ip eigrp topology`

(C) `show ip eigrp traffic`

(D) `show ip eigrp neighbors`

884. Looking at the figure below, which pathway would be used to reach the system with the IP address of 15.22.34.10?

```
R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    11.0.0.0/8 is directly connected, FastEthernet0/0
C    12.0.0.0/8 is directly connected, Serial0/3/0
D    13.0.0.0/8 [90/2172416] via 12.0.0.2, 08:19:07, Serial0/3/0
C    14.0.0.0/8 is directly connected, Serial0/3/1
D    15.0.0.0/8 [90/2681856] via 14.0.0.1, 08:19:07, Serial0/3/1
                [90/2681856] via 12.0.0.2, 08:19:07, Serial0/3/0
D    16.0.0.0/8 [90/2172416] via 14.0.0.1, 08:19:07, Serial0/3/1
R1#
```

(A) R1 would always send to 14.0.0.1 unless it fails.

(B) R1 would alternate between 14.0.0.1 and 12.0.0.2.

(C) R1 would always send to 12.0.0.2 unless it fails.

(D) R1 would send to 13.0.0.1 unless it fails.

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885. You have configured EIGRP and wish to check the queue count and retransmit interval for established adjacency. What command would you use?

(A) `show ip eigrp neighbors`

(B) `show ip eigrp topology`

(C) `show ip eigrp traffic`

(D) `show ip eigrp interfaces`

886. Looking at the figure below, how many paths can EIGRP use to the same destination?

```
R1#show ip protocols

Routing Protocol is "eigrp  10 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
  EIGRP maximum hopcount 100
  EIGRP maximum metric variance 1
Redistributing: eigrp 10
  Automatic network summarization is in effect
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
     11.0.0.0
     12.0.0.0
  Routing Information Sources:
    Gateway         Distance      Last Update
    12.0.0.2        90            99622
  Distance: internal 90 external 170
```

(A) 4

(B) 10

(C) 90

(D) 170

887. You are troubleshooting EIGRP. What commands could you use to view the autonomous system number? (Select two.)

(A) `show ip protocols`

(B) `show ip route`

(C) `show ip eigrp as_num`

(D) `show ip eigrp topology`

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888. Looking at the figure below, how long does it take router R1 to send an EIGRP packet to its neighbor and receive a reply?

```
R1#show ip eigrp neighbors
IP-EIGRP neighbors for process 10
H   Address         Interface      Hold Uptime    SRTT   RTO   Q   Seq
                                   (sec)          (ms)        Cnt  Num
0   12.0.0.2        Se0/3/0        10   00:03:11  40     1000  0   4
```

(A) 10 seconds

(B) 0 seconds

(C) 4 ms

(D) 40 ms

889. What command was used in the figure below to display the output?

```
Routing Protocol is "eigrp  10 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP metric weight K1=1, K2=0, K3=1, K4=0, K5=0
  EIGRP maximum hopcount 100
  EIGRP maximum metric variance 1
Redistributing: eigrp 10
  Automatic network summarization is in effect
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
     11.0.0.0
     12.0.0.0
  Routing Information Sources:
    Gateway         Distance      Last Update
    12.0.0.2        90            99622
  Distance: internal 90 external 170
```

(A) `show ip protocols`

(B) `show ip route`

(C) `show ip eigrp neighbors`

(D) `show ip eigrp topology`